

Rhea Madhogarhia

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EDUCATION

The University of Chicago | Chicago, IL

BS Computer Science, BA Cognitive Science | Expected Graduation: June 2026 | **GPA:** 3.76/4.0

Awards: Quad Faculty Research Grant (AY 2024-25), Jeff Metcalf Fellowship (2023)

TECHNICAL SKILLS

Programming Languages: Python, Java, Go, C, SQL, JavaScript, HTML/CSS. **Frameworks & Libraries:** Svelte, Pandas, NLTK, Vega-lite. **Developer Tools:** Git, GitHub, BigQuery, Airflow, Google Colab, Bash, Vim, Documentation Writing

WORK EXPERIENCE

Machine Learning Engineering Intern | Reddit

New York, NY | Summer 2025

- Engineered a novel First Pass Ranker (FPR) in Go and Python, using user engagement signals to improve search result relevance.
- Implemented and tuned (Bayesian Optimization) a ratio-driven boosting algorithm to prioritize high-value content and penalize clickbait, enhancing search candidate quality.
- Authored the formal design document for the "Signals FPR" and managed the full project lifecycle from implementation to launching user-facing A/B tests

Research Assistant | Cognition, Attention, and Brain Lab

Chicago, IL | May 2024 – Present

- Conduct data processing & coding, ML-assisted transcription of fMRI audio data, and NLP sentiment analysis
- Co-authored study titled "Comprehension of causal structure in narratives" (**Principal Investigator:** Monica Rosenberg)

Data Science & ML Intern | Charactour

Chicago, IL | Summer 2024

- Performed exploratory data analysis on personality traits from 9M+ users using Python (Pandas).
- Used BERT Regression to discover correlations between media properties and their fanbases.
- Preprocessed text data with NLTK and built predictive models (MLP & Elastic Net) using user and media trait datasets.

Machine Learning & Data Intern | Blue Fever

Los Angeles, CA | Summer 2023

- Drove development of a new program to automatically reply to customers, leading to increased user engagement using prompt engineering, Python, and SQL (BigQuery)
 - Leveraged prior experience as help-line supporter to guide bot responses, prioritizing authenticity and sensitivity towards youth audience
 - Collaborated with C-Suite leaders to design voice, image, and brand of response bot
 - Conducted sentiment analysis on user data, presenting key product insights to C-Suite leadership
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ACHIEVEMENTS & RECOGNITION

The Rithm Project: Invited attendee for the 2024 'Human Connection + AI Summit'.

Summer Tech Showcase 2024: Grant Recipient & People's Choice Award winner for building a digital archive platform for the UChicago Crew team.

Young Futures Funding Challenge: Selected top 30 finalists from 100+ proposals for the "Under Pressure" Youth Tech & Wellbeing challenge.

ACTIVITIES & INTERESTS

A Cappella Vocalist (Voices In Your Head), Improvisational Comedy ([Off-Off Campus](#)), Sketch & Play Writing, Talk-Show Creator and Co-Host ([Finders, Keepers](#) for WHPK 88.5fm), UChicago Bhangra Team, Support help-line Supporter and Training Coordinator, [Amoofy](#) (start-up company) Advisor & Web Dev, TEAL Certificate, Audio Editing (Logic), Data Cognition Lab - Alex Kale,